fsoft2026 1DD 1: Iteration 1 – Technical Report Domain Model, Glossary, Functional and Non-Functional Requirements, System Sequence Diagram

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Abstract. In the following pages, the reader will be able to be informed about the application's domain model, glossary, functional and non-functional requirements, containing use cases and their specifications, and system sequence diagrams (SSD) for the MedManager project.

Keywords: Business, C++, Vaccination Center, MedManager, Management, Software Development, Command-line Interface, Health

Table of Contents

fsoft 2026 1DD 1: Iteration 1 – Technical Report 1
A. Moreira, G. Oliveira and P. Guilherme
1 Introduction 3
2 Domain Model 4
3 Glossary 5
4 Non-Functional Requirements 6
4.1 Usability 6
4.2 Reliability 6
4.3 Performance 6
4.4 Supportability 6
4.5 Design Constraints 6
4.6 Implementation Constraints 6
4.7 Interface Constraints 6
5 Functional Requirements 7
5.1 Actor: Center Administrator 7

1 Introduction

The MedManager project is a stand-alone management system designed for a single local vaccination center. It focuses on the control of physical vaccine stock, appointment scheduling, and the end-to-end vaccination process. We named the application being developed in this project "**MedManager**". The system's architecture incorporates four key stakeholder interfaces, each serving distinct but interconnected roles within the MedManager ecosystem:

- **Center Administrator:** Responsible for internal management, including specifying vaccine types, registering physical doses, and managing employee records.
- **Receptionist:** Responsible for registering SNS users, scheduling administrations, and registering patient arrivals.
- **Nurse:** Responsible for consulting the waiting room and recording the administration of vaccines.
- **SNS User:** Intends to use the application to schedule a vaccine administration.

The objective of MedManager is to provide a reliable solution that optimizes all operations related to the vaccination process at a local center. The system will be developed in C++, which is the programming language taught in this class.

2 Domain Model

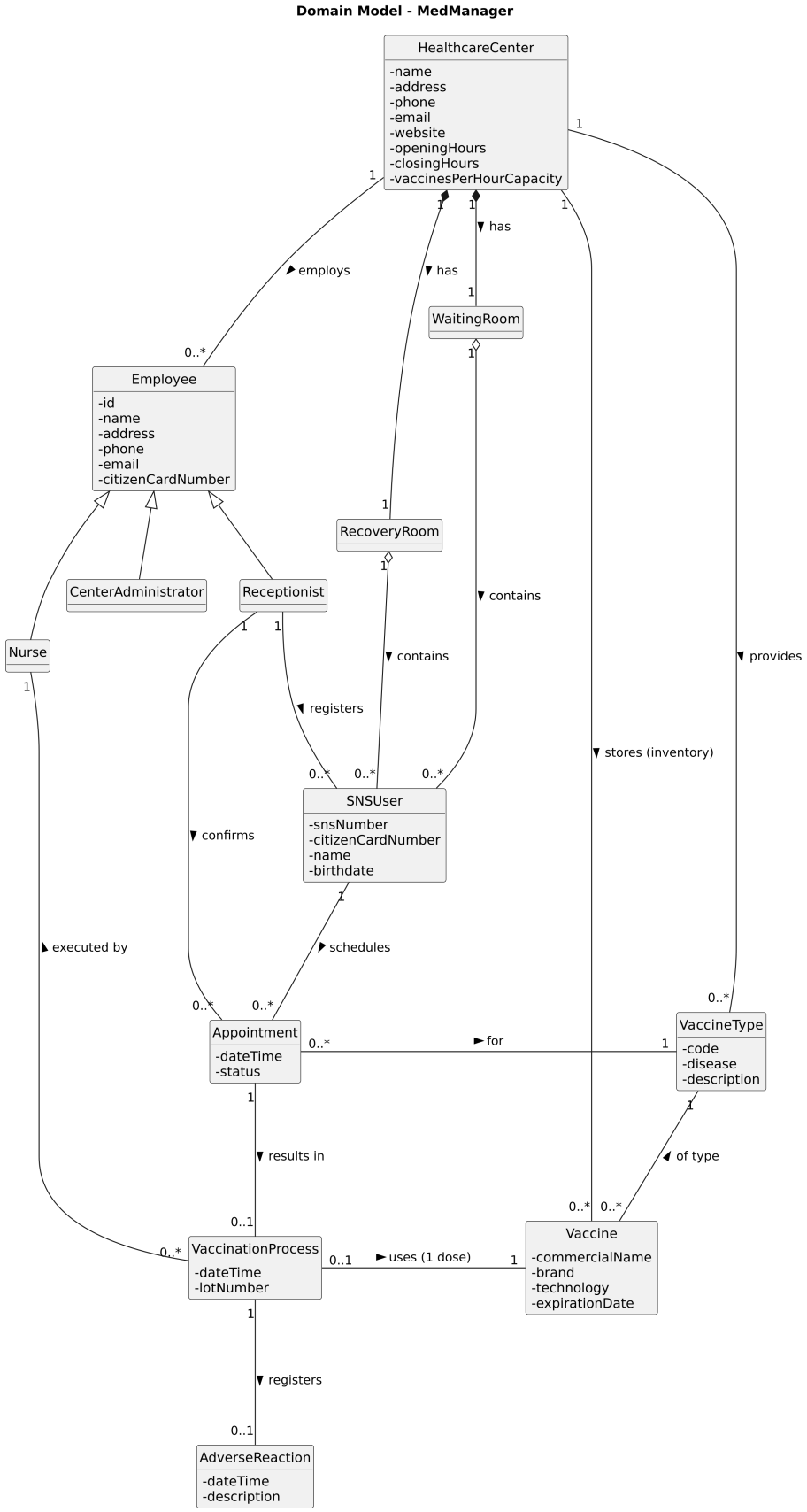

Fig. 1. MedManager's Domain Model

3 Glossary

Table 1. MedManager Glossary

Term/Expression (EN)	**Termo (PT)**	**Definition/Description**
Adverse Reaction	Reação Adversa	Clinical event or side effect recorded during the recovery period, including a description and the exact time of occurrence.
Appointment	Agendamento	A scheduled slot (date and time) for an SNS User to receive a specific vaccine type. It can be in "scheduled" or "confirmed" status.
Center Administrator	Administrador do Centro	Employee responsible for managing the center's resources, including vaccine types, physical stock, and employee records.
Employee	Funcionário	Any worker at the center, identified by ID, Name, Address, Phone, Email, and Citizen Card Number.
Healthcare Center	Centro de Saúde	The physical facility where vaccinations occur, managing its own operating hours, hourly capacity, and staff.
Nurse	Enfermeiro	Healthcare professional responsible for the clinical act of vaccination and monitoring users in the recovery room.
Receptionist	Rececionista	Employee who manages user registration and arrival, moving patients to the waiting room.
Recovery Room	Sala de Recuperação	A monitored area where users must remain for a specific period after vaccination to check for adverse reactions.
SNS User	Utente do SNS	A citizen identified by a unique SNS number, CC number, and birthdate, eligible for vaccination.
Vaccination Process	Processo de Vacinação	The formal record of a vaccine dose administration, linking a nurse, a user, and recording the specific lot number used.
Vaccine	Vacina	A physical unit in the inventory, characterized by its commercial name, brand, technology, and expiration date.
Vaccine Type	Tipo de Vacina	A category defined by a code and the disease it prevents (e.g., COVID-19). It serves as a template for physical vaccines.
Waiting Room	Sala de Espera	A logical queue (First-Come, First-Served) where users stay after check-in until a nurse is available.

4 Non-Functional Requirements

4.1 Usability

- Command-line based user interface
- Informative, yet simple, error messages
- Easy-to-use option menus

4.2 Reliability

- Data storage must be reliable
- System must be available at any time
- Validation of all user inputs, including Portuguese-format phone numbers and citizen card numbers

4.3 Performance

- Response time under 1 second for all operations

4.4 Supportability

- Full application support through GitHub repository
- Recurrent application testing

4.5 Design Constraints

- Data persistent and kept in binary files
- No duplicate entries for the same SNS user on the same day

4.6 Implementation Constraints

- C++ programming language

4.7 Interface Constraints

- No graphical interface; interaction is exclusively via command-line

5 Functional Requirements

5.1 Actor: Center Administrator

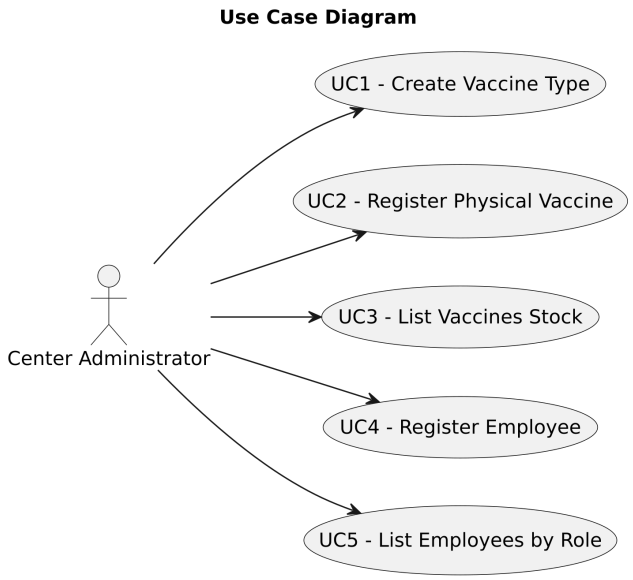

Fig. 2. Center Administrator's Use Case Diagram

Use Case 1: Specify New Vaccine Type (Specification)

Table 2. Use Case 1: Create Vaccine Type

Actor	Center Administrator
Use case name	Create New Vaccine Type
Description	The administrator creates a new category of vaccine in the system's catalog.
Precondition	The user must be logged in as Center Administrator.
Post-condition	A new Vaccine Type is registered and available for stock management.
Main flow	1. User starts the creation of a new vaccine type. 2. System requests the necessary data (code, disease, description). 3. User inputs the requested data. 4. System shows the inputted data and requests confirmation. 5. User confirms the data. 6. System registers the new vaccine type and informs operation success.
Alternative path	If the code already exists or data format is invalid, the system informs the user and resumes at step 2.

SSD: UC1 – Create Vaccine Type

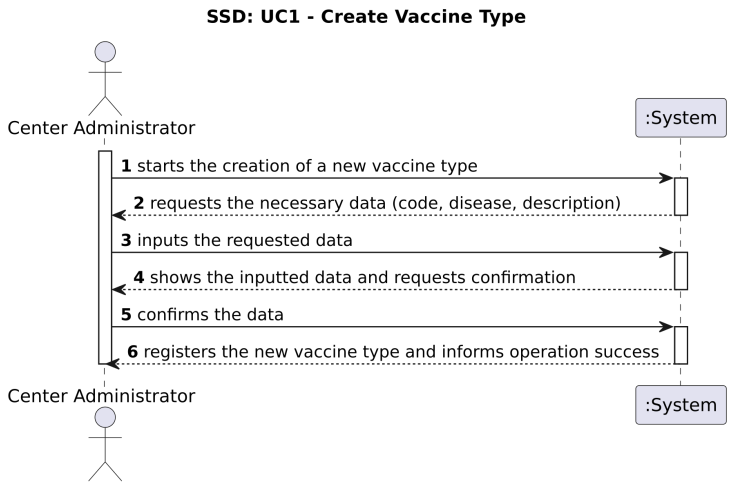

Fig. 3. SSD: UC1 – Create Vaccine Type

Use Case 2: Register Physical Vaccine Dose (Specification)

Table 3. Use Case 2: Register Physical Vaccine

Actor	Center Administrator
Use case name	Register Physical Vaccine
Description	The administrator adds physical vaccine units to the center's inventory.
Precondition	At least one Vaccine Type must exist in the system.
Post-condition	Physical doses are registered in the system and associated with a Vaccine Type.
Main flow	1. User starts the registration of a physical vaccine. 2. System shows the list of available Vaccine Types . 3. User selects the corresponding type. 4. System shows a list of vaccine technologies (mRNA, Viral Vector, etc.). 5. User selects the technology. 6. System requests details (Brand, Lot Number, Expiration Date). 7. User inputs the requested data. 8. System shows the data and requests confirmation. 9. User confirms the data. 10. System registers the vaccine doses and informs operation success.
Alternative path	If no Vaccine Types exist, the system informs the user and the operation is aborted.

SSD: UC2 – Register Physical Vaccine Dose

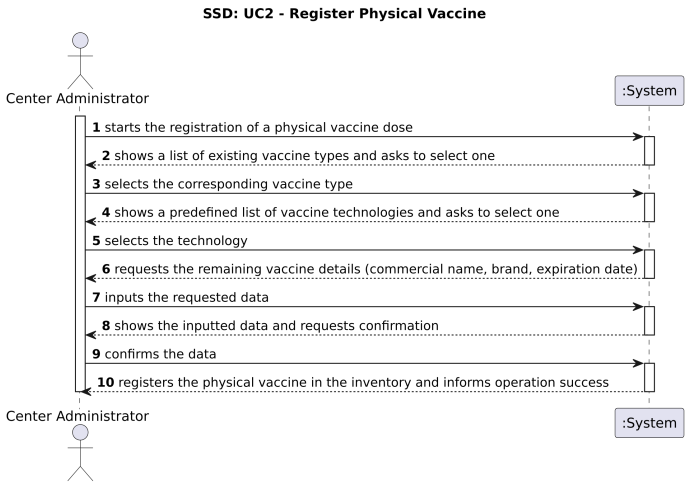

Fig. 4. SSD: UC2 – Register Physical Vaccine Dose

Use Case 3: List of Vaccine Stock (Specification)

Table 4. Use Case 3: List of Vaccine Stock

Actor	Center Administrator
Use case name	List Vaccine Stock
Description	The administrator consults the current physical vaccine doses available in the center.
Precondition	At least one vaccine dose must be registered in the system.
Post-condition	The system displays the inventory grouped by type and brand.
Main flow	1. User starts the request to list the vaccine stock. 2. System checks if there are vaccines registered in the center. 3. System displays a list of all vaccine doses grouped by type and brand.
Alternative path	2. If no vaccines are registered, the system informs the user and returns to the main menu.

SSD: UC3 – List Vaccine Stock

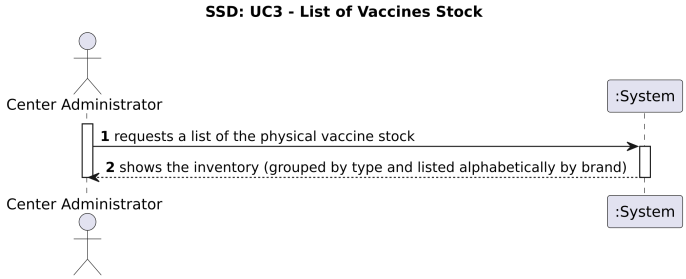

Fig. 5. SSD: UC3 – List Vaccine Stock

Use Case 4: Register New Employee (Specification)

Table 5. Use Case 4: Register New Employee

Actor	Center Administrator
Use case name	Register New Employee
Description	The administrator adds a new staff member (Nurse, Receptionist) to the system.
Precondition	The user must be logged in as Center Administrator.
Post-condition	A new employee record is created and stored in the staff log.
Main flow	1. User starts the registration of a new employee. 2. System shows the available roles (Nurse, Receptionist). 3. User selects the intended role. 4. System requests personal data (Id, Name, Address, Phone, Email, CC). 5. User inputs the requested data. 6. System shows the data and requests confirmation. 7. User confirms the data. 8. System validates data formats and uniqueness, then registers the employee.
Alternative path	If the CC, Email, or Phone is already in use, the system informs the user and resumes at step 4.

SSD: UC4 – Register New Employee

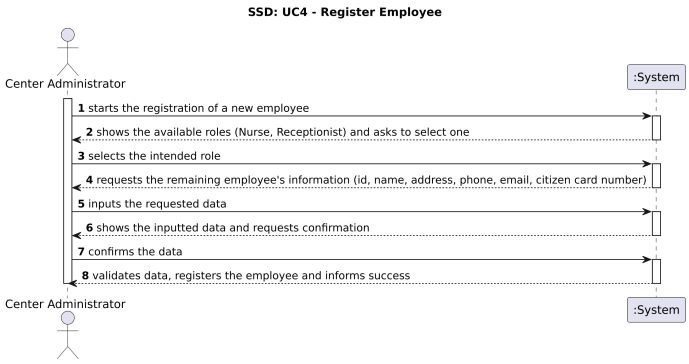

Fig. 6. SSD: UC4 – Register New Employee

Use Case 5: List Employees by Role (Specification)

Table 6. Use Case 5: List Employees by Role

Actor	Center Administrator
Use case name	List Employees by Role
Description	The administrator consults the staff members filtered by their professional role.
Precondition	Employees must be registered in the system.
Post-condition	The system displays the list of employees for the selected role.
Main flow	1. User starts the request to list employees by role. 2. System shows the available roles. 3. User selects the specific role. 4. System displays all employees registered under the selected role.
Alternative path	If no employees exist for that role, the system informs the user.

SSD: UC5 – List Employees by Role

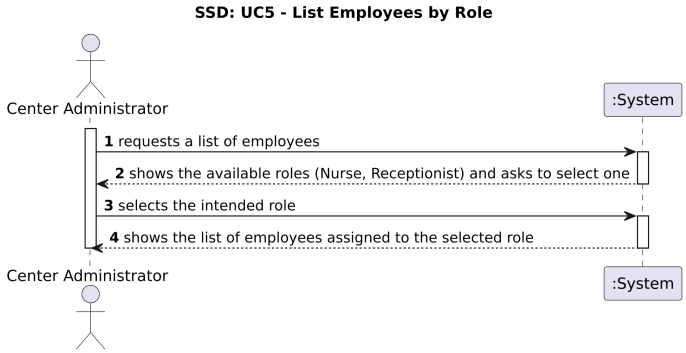

Fig. 7. SSD: UC5 – List Employees by Role